

1. An ideal transformer with purely resistive load operates at 12 kV on the primary side. It supplies electrical energy to a number of nearby houses at 120 V. The average rate of energy consumption in the houses served by the transformer is 60 kW. The value of resistive load (R_s) required in the secondary circuit will be _____ m Ω .

[2023 (06 Apr Shift 1)]

2. A capacitor of capacitance 150.0 μF is connected to an alternating source of emf given by $E = 36 \sin(120\pi t)$ V. The maximum value of current in the circuit is approximately equal to:

[2023 (06 Apr Shift 2)]

- (1) 2 A
(2) $\sqrt{2}$ A
(3) $2\sqrt{2}$ A
(4) $\frac{1}{\sqrt{2}}$ A

3. An oscillating LC circuit consists of a 75 mH inductor and a 1.2 μF capacitor. If the maximum charge to the capacitor is 2.7 μC . The maximum current in the circuit will be _____ mA.

[2023 (08 Apr Shift 1)]

4. A series combination of resistor of resistance 100 Ω inductor of inductance 1 H and capacitor of capacitance 6.25 μF is connected to an ac source. The quality factor of the circuit will be _____.

[2023 (08 Apr Shift 2)]

5. Given below are two statements:

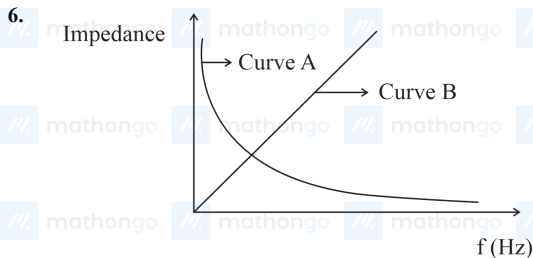
Statement I: Maximum power is dissipated in a circuit containing an inductor, a capacitor and a resistor connected in series with an AC source, when resonance occurs.

Statement II: Maximum power is dissipated in a circuit containing pure resistor due to zero phase difference between current and voltage.

In the light of the above statements, choose the correct answer from the options given below:

[2023 (10 Apr Shift 1)]

- (1) Statement I is false but Statement II is true
(2) Both Statement I and Statement II are true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false



As per the given graph, choose the correct representation for curve A and curve B

{Where X_C = Reactance of pure capacitive circuit connected with A.C. source

X_L = Reactance of pure inductive circuit connected with A.C. source

R = Impedance of pure resistive circuit connected with A.C. source

Z = Impedance of the LCR series circuit }

[2023 (11 Apr Shift 1)]

- (1) $A = X_C, B = R$
(2) $A = X_L, B = R$
(3) $A = X_L, B = Z$
(4) $A = X_C, B = X_L$

7. Given below are two statements:

Statement I : When the frequency of an AC source in a series LCR circuit increases, the current in the circuit first increases, attains a maximum value and then decreases.

Statement II : In a series LCR circuit, the value of power factor at resonance is one.

In the light of given statements, choose the most appropriate answer from the options given below.

[2023 (12 Apr Shift 1)]

- (1) Statement I is incorrect but Statement II is true
- (2) Both Statement I and Statement II are false
- (3) Both Statement I and Statement II are true
- (4) Statement I is correct but Statement II is false

8. Given below are two statements:

Statement I : An AC circuit undergoes electrical resonance if it contains either a capacitor or an inductor.

Statement II: An AC circuit containing a pure capacitor or a pure inductor consumes high power due to its non-zero power factor. In the light of above statements, choose the correct answer from the options given below:

[2023 (13 Apr Shift 2)]

- (1) Statement I is false but statement II is true
- (2) Statement I is true but statement II is false
- (3) Both Statement I and Statement II are false
- (4) Both Statement I and Statement II are true

ANSWER KEYS

1. (240) 2. (1) 3. (9) 4. (4) 5. (2) 6. (4) 7. (3) 8. (3)

1. (240)

The given data is

$$P = 60 \times 10^3 \text{ W}$$

$$V_s = 120 \text{ V}$$

A transformer has a primary coil and a secondary coil.

The formula for the power delivered is given by

$$P = \frac{V_s^2}{R_s}$$

$$\Rightarrow 60 \times 10^3 = \frac{120 \times 120}{R_s}$$

$$\Rightarrow R_s = \frac{120 \times 120}{60 \times 10^3} = 240 \text{ m}\Omega$$

2. (1)

The formula for maximum current is given by

$$I_{\max} = \omega C V_m \dots (i)$$

The given data is

$$\omega = 120\pi$$

$$V_m = 36 \text{ V}$$

$$C = 150 \mu\text{F}$$

Substituting the values in equation (i)

$$I_{\max} = (120\pi) \times (150 \times 10^{-6}) \times 36$$

$$= 2.036 \text{ A}$$

$$\approx 2 \text{ A}$$

3. (9)

When the capacitor is fully discharged, the current is maximum and all the energy resides in the inductor. Hence, the energy of the capacitor is equal to the energy of the inductor.

$$\frac{L i_m^2}{2} = \frac{Q_m^2}{2C} \dots (i)$$

where i_m is the maximum current and Q_m is the maximum charge.

The given values are

$$Q_m = 2.7 \times 10^{-6} \text{ C}$$

$$L = 75 \times 10^{-3} \text{ H}$$

$$C = 1.2 \times 10^{-6} \text{ F}$$

The maximum current is found from equation (i)

$$i_m = \sqrt{\frac{1}{LC}} Q_m$$

$$\Rightarrow i_m = \frac{2.7 \times 10^{-6}}{\sqrt{75 \times 10^{-3} \times 1.2 \times 10^{-6}}} = 9 \text{ mA}$$

4. (4)

The formula to calculate the quality factor of an LCR circuit is given by

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} \dots (1)$$

Substitute the values of the known parameters into equation (1) to calculate the required quality factor of the circuit.

$$Q = \frac{1}{100} \sqrt{\frac{1}{6.25 \times 10^{-6}}}$$

$$= \frac{1}{100} \times \frac{1000}{2.5}$$

$$= 4$$

5. (2)

The formula for impedance is given by

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

At resonance, $X_L = X_C$. So, $Z_{\min} = R$

Power will be maximum when impedance is minimum. Thus the maximum power occurs in a purely resistive circuit.

6. (4)

The formula to calculate the inductive reactance is

$$X_L = \omega L \dots (1)$$

The formula to calculate capacitive reactance is

$$X_C = \left(\frac{1}{\omega C}\right) \dots (2)$$

As can be seen from equation (1), the inductive reactance varies directly with the frequency and equation (2) indicates that capacitive reactance varies inversely with the frequency.

Hence, curve A represents the capacitive reactance and curve B represents the inductive reactance.

7. (3)

When an a.c. voltage is applied in a series LCR circuit, the current is increased first, attains a maximum and then decreases.

The formula to calculate the power factor is given by

$$\cos \phi = \frac{R}{\sqrt{R^2 + (X_L^2 - X_C^2)}} \dots (1)$$

At resonance, $X_L = X_C$, Hence,

$$\cos \phi_{\text{resonance}} = 1$$

Thus, both the statements are true.

8. (3)

For resonance to occur the condition is $X_L = X_C$.

So, both inductor and capacitor is required.

The power factor is given by $\cos \phi$.

For an inductor the power factor is zero-lagging, $\cos 90^\circ = 0$. The angle between the voltage and current is 90° .

For a capacitor the power factor is zero-leading.

Hence, both statements are false.